

Tests on running inference on the mediapipe .tflite obd model inference tool web for l3f

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Situation overview

This document is a summary of tests run on the mediapipe .tflite obd model inference web tool I built for the l3f project. It aims to confirm the web tool is reliable & usable across multiple hardware options without crashing & misbehavior. This is a summary of multiple tests

Setup

Tests For every test we record the following environment variables:

- device
- OS
- browser + version
- camera
- target sampling rate
- model
- delegate
- score threshold
- goal of test
- duration of test
- special notes on env

, where

- **browser + version** is (index.html -> Environment -> Browser) rendering
- **target sampling rate** is the fps parameter given to `startCameraInferenceLoop()` when called from within `handleStartCamera()` in `src/main.js`
- **model** is: DIRNAME where `data_and_training/models/DIRNAME/model.tflite` is identical to `browser_app/public/models/model.tflite`; Test: `cmp browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite -> silent exit, no browser_app/public/models/model.tflite data_and_training/models/DIRNAME/model.tflite differ: byte xx, line y type output.`
- **delegate** is the `delegate` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`
- **score threshold** is the `scoreThreshold` parameter given to `ObjectDetector.createFromOptions` in model loading/compilation section of `try` block of `startApplication()` function in `src/main.js`

and we collect

- csv formatted measurements (camera fpc, sampling fps, inference fps, inference latency) & appropriate visualisations
- UI behavior observations
- browser command line prints of note
- other notes

, where

browser command line prints of note means we ignore per-detection/per-measurement prints, but note any execution time errors.

Technical Render this document by running `Rscript -e "rmarkdown::render('reporting.Rmd')"` From `browser_app/measurements/`

Libraries:

```
# For data manipulation
suppressPackageStartupMessages(library(dplyr)) # <- Don't print warning on import
library(tidyr)

# For plotting & rendering
library(ggplot2)
library(scales)
```

Diagram drawing function (edited for Rmd use from `data_and_training/devtools/fpsdiagram.R`):

```
fpsdiagram <- function(input_filename) {

  if (!file.exists(input_filename)) {
    stop(
      paste0("Error: File '", input_filename, "' not found."),
      call. = FALSE
    )
  }

  df <- read.csv(input_filename, stringsAsFactors = FALSE)

  df$Timestamp <- as.POSIXct(
    df$Timestamp,
    format = "%Y-%m-%dT%H:%M:%OSZ",
    tz = "UTC"
  )

  df_long <- df %>%
    pivot_longer(
      cols = c(Camera_FPS, Sampling_FPS, Inference_FPS),
      names_to = "Metric",
      values_to = "FPS"
    )

  p <- ggplot(
    df_long,
    aes(x = Timestamp, y = FPS, color = Metric)
  ) +
    geom_line(linewidth = 1) +
    geom_point(size = 2) +
    scale_x_datetime(labels = date_format("%H:%M:%S")) +
    labs(
      title = "Performance Monitoring Metrics",
      subtitle = paste("File:", input_filename),
      x = "Timestamp (UTC)",
      y = "FPS",
      color = "Metric"
    ) +
```

```

theme_minimal(base_size = 12) +
theme(
  legend.position = "bottom",
  axis.text.x = element_text(angle = 45, hjust = 1)
)

print(p)
}

```

Test series 1: Finding the optimal sampling rate

Test 1.1 starting from a low baseline

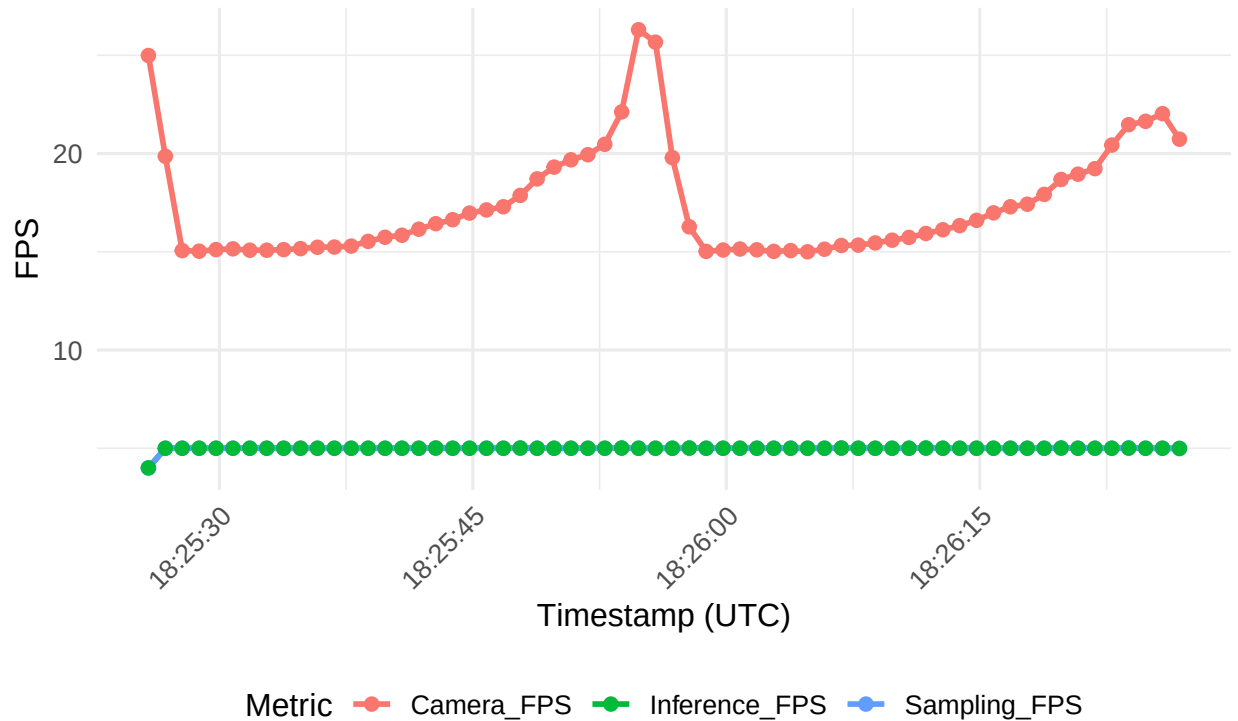
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 5
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr5_20260921_212518.csv")
```

Performance Monitoring Metrics

File: fps_sr5_20260921_212518.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, significantly below Camera_FPS, can raise.

Test 1.2 raising to 10 fps sampling

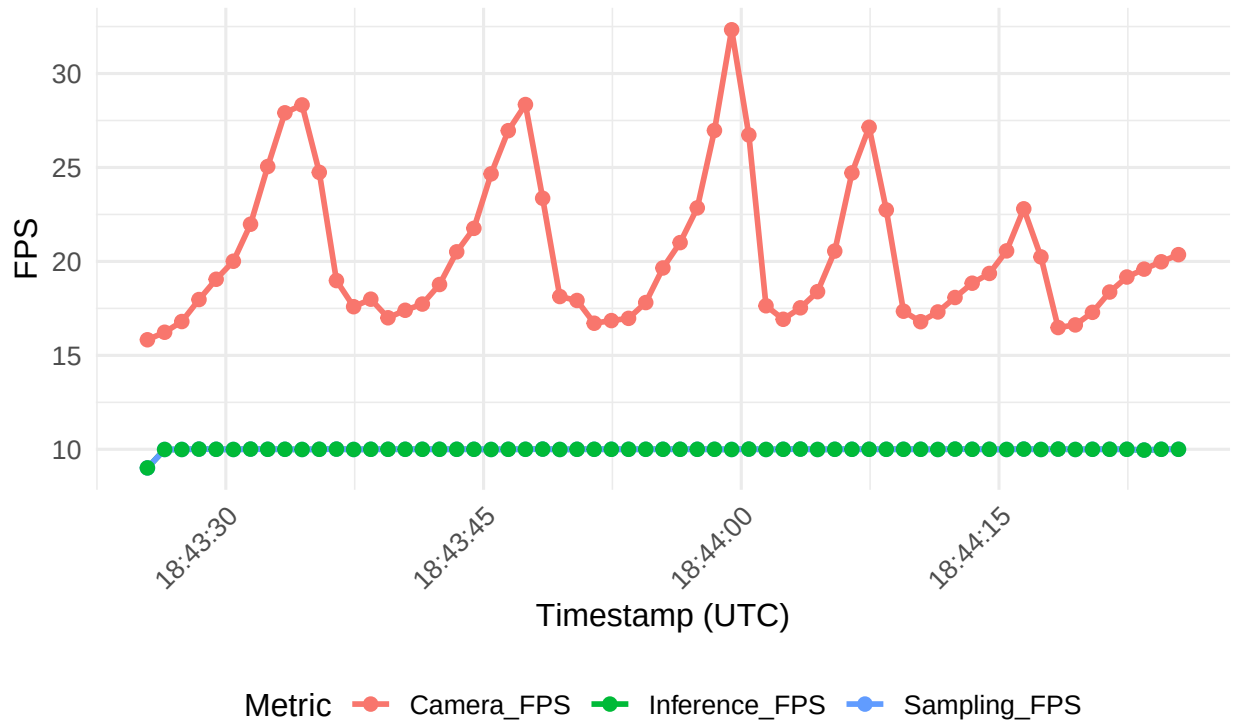
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 10
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr10_20260921_214321.csv")
```

Performance Monitoring Metrics

File: fps_sr10_20260921_214321.csv



Results

No record of browser command line prints.

Notes: Inference_FPS matches Sampling_FPS, still below Camera_FPS, can raise.

Test 1.3 reaching camera fps

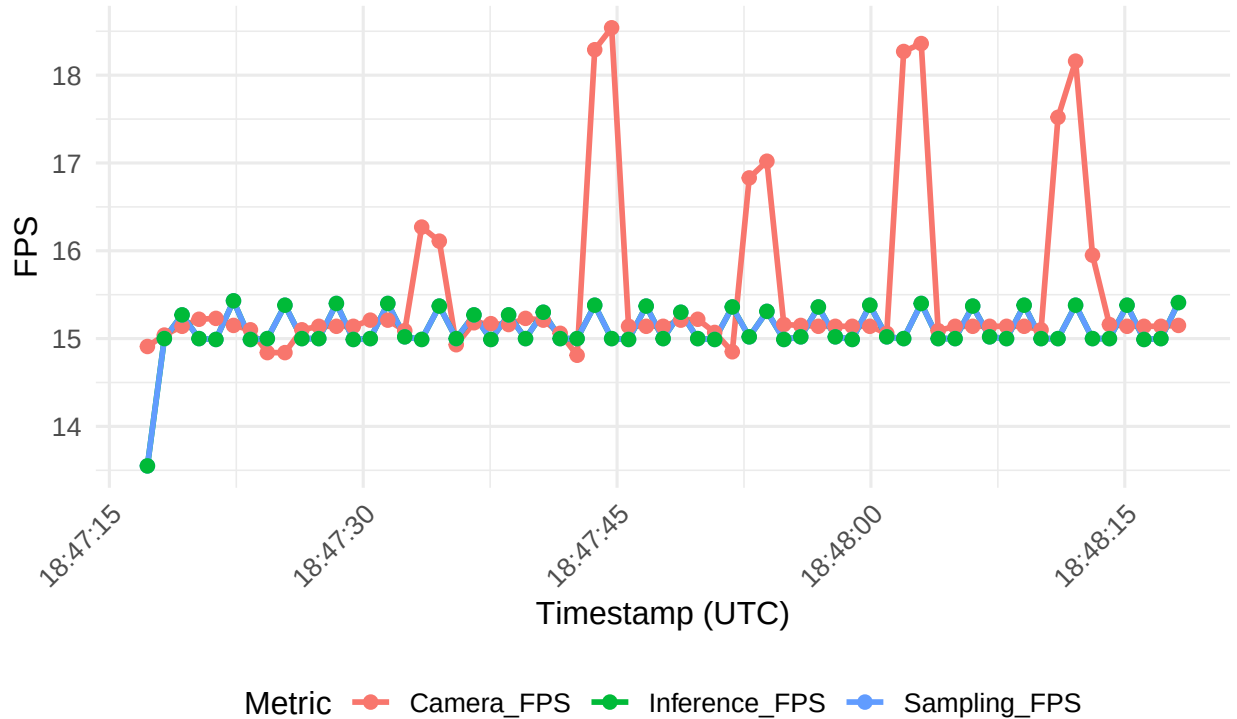
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 15
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr15_20260921_214710.csv")
```

Performance Monitoring Metrics

File: fps_sr15_20260921_214710.csv



Results

No record of browser command line prints.

Notes: Inference_FPS & Sampling_FPS now occasionally above Camera_FPS, will lower.

Test 1.4 achieving stability

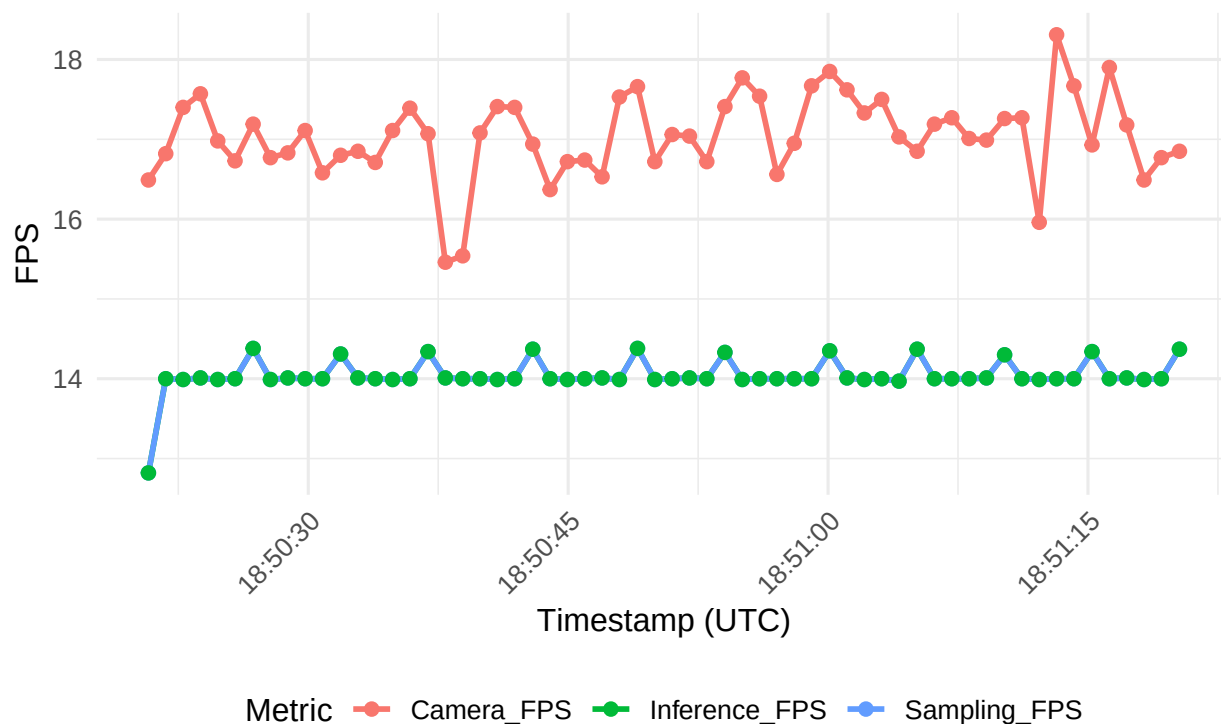
Environment

- device: Self built PC (AMD Ryzen 7 3700X, 16GB RAM, Nvidia Geforce GTX 1660 Super)
- OS: Fedora Linux 42 (Workstation Edition)
- browser + version: Mozilla/5.0 (X11; Linux x86_64; rv:143.0) Gecko/20100101 Firefox/143.0
- camera: Logitech, Inc. C920 HD Pro Webcam
- target sampling rate: 14
- model: `ssd_plus_mobilenet_v2_13f_20260807_225848`
- delegate: CPU
- score treshold: 0.35
- goal of test: baseline establishment
- duration of test: ~1 min
- special notes on env: -

```
fpsdiagram("fps_sr14_20260921_215016.csv")
```

Performance Monitoring Metrics

File: fps_sr14_20260921_215016.csv



Results

No record of browser command line prints.

Notes: **Inference_FPS** & **Sampling_FPS** now stable below **Camera_FPS**, not surpassing it. NOTE: **Latency_MS** (how long it actually takes to complete an inference) ranges from 46 to 49. It is still realistic to raise it. We're bottlenecked by **Camera_FPS** on this hardware.